

GRADE: 8	NAME: Proportional Reasoning – 2
LESSON NO. : 2	
No. of Sessions/Periods	10
Lesson Overview	What is the focus of the Unit? The focus of this unit is extending the concept of proportionality to ratios with more than two terms, understanding Representative Fractions (RF) in maps, and distinguishing between direct and inverse proportions in real-world contexts.
Previous Knowledge	What prior Knowledge the students need in order to be successful with this unit? Students need to understand basic ratios (two terms), cross-multiplication for verification, and the concept of direct proportion where quantities change by the same factor.
Vocabulary /Terminology	What is the new vocabulary/ terminology used in this lesson? Proportionality, Representative Fraction (RF), Multi-term Ratio, Simplest Form, Direct Proportion, Inverse Proportion, Constant of Proportionality (k), Pie Chart
SESSION NO. : 01	CONCEPT NAME: Recap of Proportionality Ratios and Cross-Multiplication Recap
Learning Outcomes	Students will be able to verify if two ratios are proportional using cross-multiplication.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Textbook (Page 55), whiteboard, markers.
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment
Time Assigned: 10 minutes	Use the "Idli Batter" example. If 2 cups of rice need 1 cup of dal (2:1), how much dal for 6 cups of rice? Methodology: Demonstrate the cross-multiplication method ($a \times d = b \times c$) to verify proportionality.
Student Activity	Solve the Viswanath vs. Puneet idli mixture problem to see if they taste the same
Time Assigned: 25 minutes	
Assessment	Quick quiz on verifying ratios like 6:3 and 4:2
Assignment/HW	Find two real-life examples of ratios used in your kitchen.
Skills/Competencies	Calculation, Verification
Intelligences Addressed	Logical-Mathematical.
SESSION NO. : 02	CONCEPT NAME: Ratios in Maps Representative Fraction (RF)
Learning Outcomes	Students will understand what RF 1:60,00,000 indicates on a map.

Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Atlas, rulers, Map of South India (Page 56)
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment Show the ratio in the corner of a map. Ask: "What does 1:60,00,000 mean?"
Time Assigned: 10 minutes	Methodology: Explain that RF is Map Distance : Actual Ground Distance.
Student Activity	Convert 60,00,000 cm to km (Answer: 60 km). Use a ruler to find the distance between Bengaluru and Chennai on the map.
Time Assigned: 25 minutes	
Assessment	Calculate actual distance if map distance is 5 cm and RF is 1:50,000.
Assignment/HW	Find the RF on any map in your house or newspaper.
Skills/Competencies	Measurement, Scale Conversion.
Intelligences Addressed	Visual-Spatial.
SESSION NO. : 03	CONCEPT NAME: Map Making Activity Practical Application of Scale
Learning Outcomes	Students will create a scaled drawing of their classroom.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Measuring tape, chart paper, pencils.
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment Set the scale for the classroom map at 1:50.
Time Assigned: 10 minutes	Methodology: Guide students on how to represent large objects (blackboard, desks) using symbols.
Student Activity	Measure classroom dimensions and draw them to scale on paper.
Time Assigned: 25 minutes	
Assessment	Peer review of maps for accuracy of relative positions.
Assignment/HW	Draw a rough scaled map of your study table.
Skills/Competencies	Spatial awareness, scaling.
Intelligences Addressed	Kinesthetic, Spatial.
SESSION NO. : 04	CONCEPT NAME: Ratios with More than 2 Terms Multi-term Ratios
Learning Outcomes	Students will understand that ratios can have many terms if they change by the same factor.

Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Spice mix example (Page 57).
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment Present Viswanath's spice mix ratio 8:4:2:1 (coriander:chilli:dal:methi).
Time Assigned: 10 minutes	Methodology: Show how to reduce all terms by the same factor if one ingredient changes.
Student Activity	Solve Puneet's problem: if he has only 2 red chillies, calculate the other ingredients (Answer: 4:2:1:0.5).
Time Assigned: 25 minutes	
Assessment	Verify if 8:4:2:1 is proportional to 4:2:1:0.5.
Assignment/HW	Create a recipe ratio for a fruit salad with 4 fruits.
Skills/Competencies	Proportional scaling.
Intelligences Addressed	Logical-Mathematical.
SESSION NO. : 05	CONCEPT NAME: Dividing a Whole in a Given Ratio Distributing Quantities
Learning Outcomes	Students will learn the formula for dividing a quantity x in ratio a:b:c.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Paint mixing example (Page 58).
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment Mixing purple paint in ratio 2:3:5 (Red:Blue:White).
Time Assigned: 10 minutes	Methodology: Explain the process: Find the sum of parts, divide the total by the sum, and multiply each part.
Student Activity	If Yasmin has 10L of white paint (5 parts), find Red and Blue needed (Answer: 4L Red, 6L Blue)
Time Assigned: 20 minutes	
Assessment	Solve Example 4: 50 ml of purple paint in 2:3:5 ratio
Assignment/HW	Solve Figure it Out Q1 (Cricket practice session).
Skills/Competencies	Algebraic application.
Intelligences Addressed	Logical-Mathematical.
SESSION NO. : 06	CONCEPT NAME: Proportions in Geometry (Triangles) Angles and Side Lengths in Ratios
Learning Outcomes	Divide 180° into angles based on a given ratio.
Resources Required – Physical, Digital,	Geometry box, protractor.

Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment Can we construct a triangle with angles in ratio 1:3:5?
Time Assigned: 10 minutes	Methodology: Calculate $\angle A, \angle B, \angle C$ using $180^\circ \times \frac{\text{part}}{\text{sum}}$.
Student Activity	Calculate the angles ($20^\circ, 60^\circ, 100^\circ$) and attempt to draw the triangle
Time Assigned: 25 minutes	
Assessment	Discuss why a triangle with side lengths 1:3:5 cannot be constructed (Triangle Inequality).
Assignment/HW	Solve Figure it Out Q4 (3:4:5 ratio sides).
Skills/Competencies	Geometrical construction.
Intelligences Addressed	Visual, Logical.
SESSION NO. : 07	CONCEPT NAME: Introduction to Pie Charts A Slice of the Pie
Learning Outcomes	Understand that pie charts show proportions of a whole where angles are proportional to data.
Resources Required - Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Circle templates, protractors.
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment Show the Grade Distribution pie chart (40 students).
Time Assigned: 10 minutes	Methodology: Explain that the total angle is 360° Use the ratio of students to find the central angle.
Student Activity	Convert the student table (A:10, B:8, C:12, D:6, E:4) into central angles.
Time Assigned: 25 minutes	
Assessment	Simplify the ratio 12:10:8:6:4 to 6:5:4:3:2.
Assignment/HW	Solve Figure it Out Q2 (TV channel percentages).
Skills/Competencies	Data Interpretation.
Intelligences Addressed	Visual, Analytical.
SESSION NO. : 08	CONCEPT NAME: Inverse Proportions – Concept Understanding Inverse Relationships
Learning Outcomes	Distinguish between direct and inverse proportions.
Resources Required - Physical, Digital,	Speed-Time table (Page 63).

Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment If you go faster (increase speed), does it take more or less time? (Inverse).
Time Assigned: 10 minutes	Methodology: Define inverse proportion as $xy = k$ (Product is constant).
Student Activity	Analyse the table: Speed 5 km/h (18 hrs) vs Speed 15 km/h (6 hrs). Verify the product is 90.
Time Assigned: 25 minutes	
Assessment	Identify if "Number of workers vs Days to complete work" is direct or inverse.
Assignment/HW	Solve Figure it Out Q1 (Identifying inverse pairs).
Skills/Competencies	Critical thinking.
Intelligences Addressed	Logical-Mathematical.
SESSION NO. : 09	CONCEPT NAME: Solving Inverse Proportion Problems The Product Rule ($x_1y_1 = x_2y_2$)
Learning Outcomes	Calculate unknown values in inverse proportion scenarios.
Resources Required – Physical, Digital, Books, Charts, PPT's, Videos, Weblinks, Worksheets etc	Practice problems (Page 66).
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment Example 5: 80 students have food for 15 days. If 20 more join, how long will food last?.
Time Assigned: 10 minutes	Methodology: Set up the equation: $80 \times 15 = 100 \times x$. Solve for x .
Student Activity	Solve the "Pump problem": 2 pumps take 18 hours, how long for 4 pumps? (Answer: 9 hours) .
Time Assigned: 25 minutes	
Assessment	Solve Figure it Out Q3 (Families and water supply)
Assignment/HW	Solve Practice Q11 (Machines and toy production).
Skills/Competencies	Problem-solving.
Intelligences Addressed	Logical.
SESSION NO. : 10	CONCEPT NAME: Mixed Practice and Summary Unit Review and Work-Together Problems
Learning Outcomes	Solve complex problems involving work-rate and multi-step proportionality.
Resources Required – Physical, Digital, Books, Charts,	Unit Summary (Page 69)

PPT's, Videos, Weblinks, Worksheets etc	
Teacher Activity	Introduction: Provocation/Recapitulation/Reassessment Example 6: Ram cuts veg in 1 hr, Shyam in 1.5 hrs. How long together?.
Time Assigned: 10 minutes	Methodology: Calculate work done in 1 hour ($\frac{1}{1} + \frac{1}{1.5} = \frac{5}{3} \text{ units}$). Use direct proportion for final time.
Student Activity	Review the Summary points. Solve Practice Q9 (School periods duration change).
Time Assigned: 25 minutes	
Assessment	Final check on differentiating $\frac{x}{y} = k$ vs $xy = n$
Assignment/HW	Complete all remaining "Figure it Out" questions from the chapter.
Skills/Competencies	Synthesis of concepts.
Intelligences Addressed	Logical, Verbal-Linguistic.